

Digital Material Passport

ID **DMP-METAL-007** - Version **1.0.0**  
Issue Date **2025-05-20** - Certificate Type **EN 10204 3.1**

| Manufacturer                                                                                                                                                                                                                                                |  | Customer                                                                                                                                                                                           | Business Transaction |                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------|
| <b>ACME Metal Works GmbH</b><br>Industrial Park 123<br>52066 Aachen<br>DE<br><a href="mailto:quality@acme-metal.example.com">quality@acme-metal.example.com</a>                                                                                             |  | <b>Bridge Constructors Europe Ltd.</b><br>Engineering Plaza 456<br>75001 Paris<br>FR<br><a href="mailto:procurement@bridgeconstructors.example.com">procurement@bridgeconstructors.example.com</a> | <b>Order</b>         | PO-98765 · <b>20000 kg</b><br>Pos. 5 · 2025-04-20                                |
|                                                                                                                                                                                                                                                             |  |                                                                                                                                                                                                    | <b>Delivery</b>      | DN-12345 · <b>20000 kg</b><br>Pos. 1 · 2025-05-19<br>Internal Order ID: SO-67890 |
| <b>Certificate Receiver</b><br><b>Civil Infrastructure Authority</b><br>Government Building 789<br>Administration Wing<br>1000 Brussels<br>BE<br><a href="mailto:inspections@infra-authority.example.gov.be">inspections@infra-authority.example.gov.be</a> |  |                                                                                                                                                                                                    |                      |                                                                                  |

Product

|                   |                              |              |                                                           |
|-------------------|------------------------------|--------------|-----------------------------------------------------------|
| Product Name      | Structural Steel S420N Plate | Name (EN)    | 1.8902                                                    |
| Batch ID          | H-45678-01                   | Shape        | Plate - Width 2500 mm - Thickness 25 mm - Length 12000 mm |
| Surface Condition | Shot blasted and primed      | Product Norm | EN 10025-3 (2019)                                         |
| Weight            | 20000 kg                     | Product Norm | EN 1090-2 (2018)                                          |
| Production Date   | 2025-05-18                   | HS Code      | 720851                                                    |
| Country of Origin | DE                           |              |                                                           |

Chemical Analysis

| Heat Number <b>H-45678</b> - Melting Process <b>EAF+LF+VD</b> - Casting Method <b>ContinuousCasting</b> - Casting Date <b>2025-05-17</b> - Sample Location <b>Ladle</b> |      |      |      |       |       |       |       |       |      |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|-------|-------|-------|-------|-------|------|--|
| Symbol                                                                                                                                                                  | C    | Mn   | Si   | P     | S     | Nb    | V     | Ti    | CEV  |  |
| Unit                                                                                                                                                                    | %    | %    | %    | %     | %     | %     | %     | %     | %    |  |
| Min                                                                                                                                                                     |      | 1.00 |      |       |       |       |       |       |      |  |
| Max                                                                                                                                                                     | 0.22 | 1.70 | 0.60 | 0.025 | 0.020 | 0.050 | 0.050 | 0.030 | 0.48 |  |
| Actual                                                                                                                                                                  | 0.16 | 1.38 | 0.32 | 0.016 | 0.008 | 0.022 | 0.034 | 0.009 | 0.40 |  |
| CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.4%                                                                                                                               |      |      |      |       |       |       |       |       |      |  |

Mechanical Properties

| Tensile Strength <sup>1</sup>                                                          | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Spec Min | Spec Max | Method        | Status |
|----------------------------------------------------------------------------------------|--------|------|-----|------|---------|------|------|----------|----------|---------------|--------|
| Value [MPa]                                                                            | Rm     | 548  | 550 | 552  | 550     | 548  | 552  | 520      | 680      | EN ISO 6892-1 | ✓      |
| Yield Strength <sup>1</sup>                                                            | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Spec Min | Spec Max | Method        | Status |
| Value [MPa]                                                                            | ReH    | 438  | 440 | 442  | 440     | 438  | 442  | 420      |          | EN ISO 6892-1 | ✓      |
| Elongation after fracture                                                              | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Spec Min | Spec Max | Method        | Status |
| 3 specimens tested, gauge length 5.65#S <sub>0</sub> , transverse to rolling direction |        |      |     |      |         |      |      |          |          |               |        |
| Value [%]                                                                              | A      | 20.5 | 21  | 21.5 | 21      | 20.5 | 21.5 | 19       |          | EN ISO 6892-1 | ✓      |
| Charpy V-notch Impact Energy                                                           | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Std Dev  | Spec Min | Method        | Status |
| 3 specimens tested at -20°C, transverse to rolling direction                           |        |      |     |      |         |      |      |          |          |               |        |
| Value [J]                                                                              | KV     | 63   | 65  | 67   | 65      | 63   | 67   | 2        | 40       | EN ISO 148-1  | ✓      |

<sup>1</sup> 3 specimens tested at 20°C, transverse to rolling direction

Supplementary Tests

| Test                             | Result / limits                                               | Method   | Status |
|----------------------------------|---------------------------------------------------------------|----------|--------|
| Ultrasonic Testing<br>Class S2E2 | Yes - No recordable indications exceeding acceptance criteria | EN 10160 | ✓      |
| Through-thickness Properties     | Z25                                                           | EN 10164 | ✓      |

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|             |                                       |                                         |   |
|-------------|---------------------------------------|-----------------------------------------|---|
| Weldability | Yes - Satisfactory welding properties | Internal Method based on EN ISO 15614-1 | ✓ |
|-------------|---------------------------------------|-----------------------------------------|---|

## Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10025-3:2019 and EN 10204:2004 type 3.1. The product complies with the Construction Products Regulation (EU) No 305/2011 and is suitable for use in structural applications according to EN 1090-2:2018, up to and including Execution Class EXC3.



Validated by **John Smith**, Quality Manager, Quality Assurance - 2025-05-20  
Validated by **Maria Schmidt**, Quality Manager, Quality Assurance - 2025-05-20